

RE(BN-142)

mw to W
 $30 \text{ dBm} = 0 \text{ dB}$
 $0 \text{ dBm} = -30 \text{ dB}$

HW1

1. 3 MHz bandwidth, 20 mbps intended capacity. SNR to achieve?

$$C = B \log_2(1 + \text{SNR})$$

$$20 \times 10^6 = 3 \times 10^6 \log_2(1 + \text{SNR})$$

for future ref. $\downarrow$ linear not dB not

$$\frac{20}{3} = \log_2(1) + \log_2(\text{SNR})$$

$$\text{SNR} = \frac{S}{N}$$

$$2^{20/3} = 1 + \text{SNR}$$

$$101.594 - 1 = \text{SNR} = 100.594 \rightarrow 10 \log_{10}(100.594) = 20.069 \text{ dB}$$

2.

receiver has received power of -134 dBm , received noise power spectral density of -174 dBm/Hz , and a BW of 2 kHz .

Max rate of error-free transmission

$$C = B \log_2(1 + \frac{S}{N})$$

$$S = -134 \text{ dBm} \rightarrow 3.981 \times 10^{-14} \text{ W}$$

$$B = 2 \text{ kHz}$$

$$N = -174 \text{ dBm/Hz} \rightarrow 3.981 \times 10^{-18} \text{ W/Hz}$$

$$= 3.981 \times 10^{-18} (2\text{K}) = 7.962 \times 10^{-15}$$

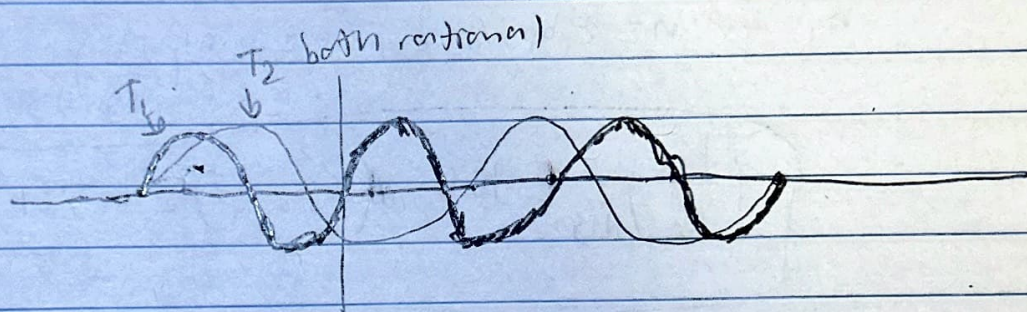
$$\frac{3.981 \times 10^{-14}}{7.962 \times 10^{-15}} = 5$$

$$C = 2000 \log_2(1 + 5)$$

$$C = 5169.925 \text{ bps}$$

3, two periodic fncs $f_1(t)$ and $f_2(t)$ w/ T_1, T_2 .
Is $f(t) = f_1(t) + f_2(t)$ always periodic?

No it is not. If either T_1 or T_2 is irrational,
there will never be a repeated output such that
 $mT_1 = nT_2$ (m and n are ints).



$3T_2 \geq 2T_1$; Never works when
 T_1 or T_2 irrat ($e, \sqrt{2}$)

4, system (digital) requires 9.6 kbps

a. If signal p/t encodes 4-bit word, min BW required?

$$R \leq 2B \log_2(M) \quad M = \# \text{ of sig levels} = 2^r = 16$$

$$R = \text{bit rate} = 9600$$

$$B = \text{BW} = ?$$

$$r = 4$$

$$9600 \leq 2B \log_2(16) \rightarrow B = \frac{9600}{8} = \boxed{1200 \text{ Hz}}$$

b. $\frac{9600}{16} = \boxed{600 \text{ Hz}}$

energy per bit

5.

b

$$C/B = \eta$$

$$\rightarrow \frac{E_b}{N_0} \geq \frac{2^{\eta}-1}{\eta} \rightarrow \left(\frac{E_b}{N_0} \right)_{dB} = 10 \log_{10} \left(\frac{2^{\eta}-1}{\eta} \right)$$

noise
at
room temp

$$(T=17^{\circ})$$

$$\approx -204 \text{ dB/Hz}$$

4. for $\eta = 2 \text{ bps/Hz}$, find $\min \left(\frac{E_b}{N_0} \right)_{dB}$

$$\left(\frac{E_b}{N_0} \right)_{dB} = 10 \log_{10} \left(\frac{2^2-1}{2} \right) = 1.761 \text{ dB}$$

$$b. \left(\frac{E_b}{N_0} \right)_{dB} = 10 \log_{10} \left(\frac{2^{\eta}-1}{\eta} \right)$$

$$\lim_{\eta \rightarrow 0} \frac{2^{\eta}-1}{\eta} = \frac{0}{0}$$

by L'Hospital rule

$$\lim_{\eta \rightarrow 0} \frac{\ln(2) 2^{\eta}}{1} = \ln(2)$$

$$\left(\frac{E_b}{N_0} \right)_{dB \min} = 10 \log_{10}(\ln(2)) = -1.572 \text{ dB}$$